

1 Natural-language Intent



“Example User Request”

“Build a low-power edge-vision SoC that captures a 1080p camera stream, runs object detection at 30 FPS, and reports results over Ethernet.”

Interactive Clarification



Power budget? → ≤5 W



Camera interface? → MIPI CSI-2

Resolve Missing or Ambiguous Fields

2 LLM Structuring & Clarification

LLM Requirement Interpreter

1 System Functions



Sensing · Communication · Application Processing
Image Capture · Object Detection · Result Reporting

2 Quantitative Constraints



Latency · Throughput · Power · Area · Memory
1080p · ≥30 FPS · ≤33 ms · ≤5 W

3 Integration Attributes



Data Types · Protocols · External Interfaces
RGB888 · MIPI CSI-2 · Ethernet

3 Machine-readable Specification



Example Specification (YAML)

```
soc_spec:
  domain: edge_vision
  functions:
    - image_capture
    - object_detection
    - result_reporting
  constraints:
    resolution: 1920x1080
    throughput_fps: ">= 30"
    latency_ms: "<= 33"
    power_w: "<= 5"
  integration:
    input_type: RGB888
    input_protocol: MIPI_CSI_2
    output_protocol: Ethernet
    output_type: detection_metadata
```



User Intent



Explicit Categories and Targets



Stable Interface for Case and IP Retrieval